

8-1.框图

GT-C6712(COLLINS) BLOCK DIAGRAM

The diagram illustrates the internal architecture of the GT-C6712(COLLINS) module, organized into three main functional blocks: Baseband, Processor, and RF/IO.

Baseband (Left Section, Light Blue):

- SKY77560:** The main baseband processor, connected to **MAIN_ANT** and **VBAT**. It interfaces with **SAW FILTER** and **DCS, PCS RX**.
- KEY:** A keypad controller connected to **KEY_ROW(0,1)** and **KEY_COL(0,1)**.
- LCD/TSP:** A liquid crystal display and touch screen processor connected to **LED+**, **LED-**, and **LCD FILTER**.
- SD CARD:** A secure digital card interface connected to **SD_DET, CMD, CLK, DATA0-3**.
- BMA220 (ACC SENSOR):** A digital accelerometer connected to **ACC_SCL** and **ACC_SDA**.
- SIM1, SIM2:** Two SIM card slots connected to **VSIM1**, **VSIM2**, **SMH1_RST, SIMMO, SIMCLK**, and **SMH1_RST, SIMMO, SIMCLK**.
- ET6302 (Dual SIM Connector):** A dual SIM connector connected to **VBAT, VDD_DIG_3.0V**, **RST, SIM, SIMCLK, SPIDAT**, **A, SPICK, SPICS, SYSRSTB**, and **SIM_IO**.
- MEMORY (2G ONE NAND/ 1G DDR SDRAM):** Connected to **VDD_MEM_1.8V**, **F_ADV_N, RST_ONE_NAND, CE_ONE_NAND, OE_N, WE_N, F_CLK, DA[0:13]**, **FADQ[0:15], DDQ[0:15]**, **DDR_CE, D_CAS_N, D_RAS_N, D_WE_N, D_CKE, D_CLK, D_CLK_N, DDQ_M_0, DDQ_S_0, DDQ_M_1, DDQ_S_1**, **N_RDY, F_RDY**.

Processor (Middle Section, Light Green):

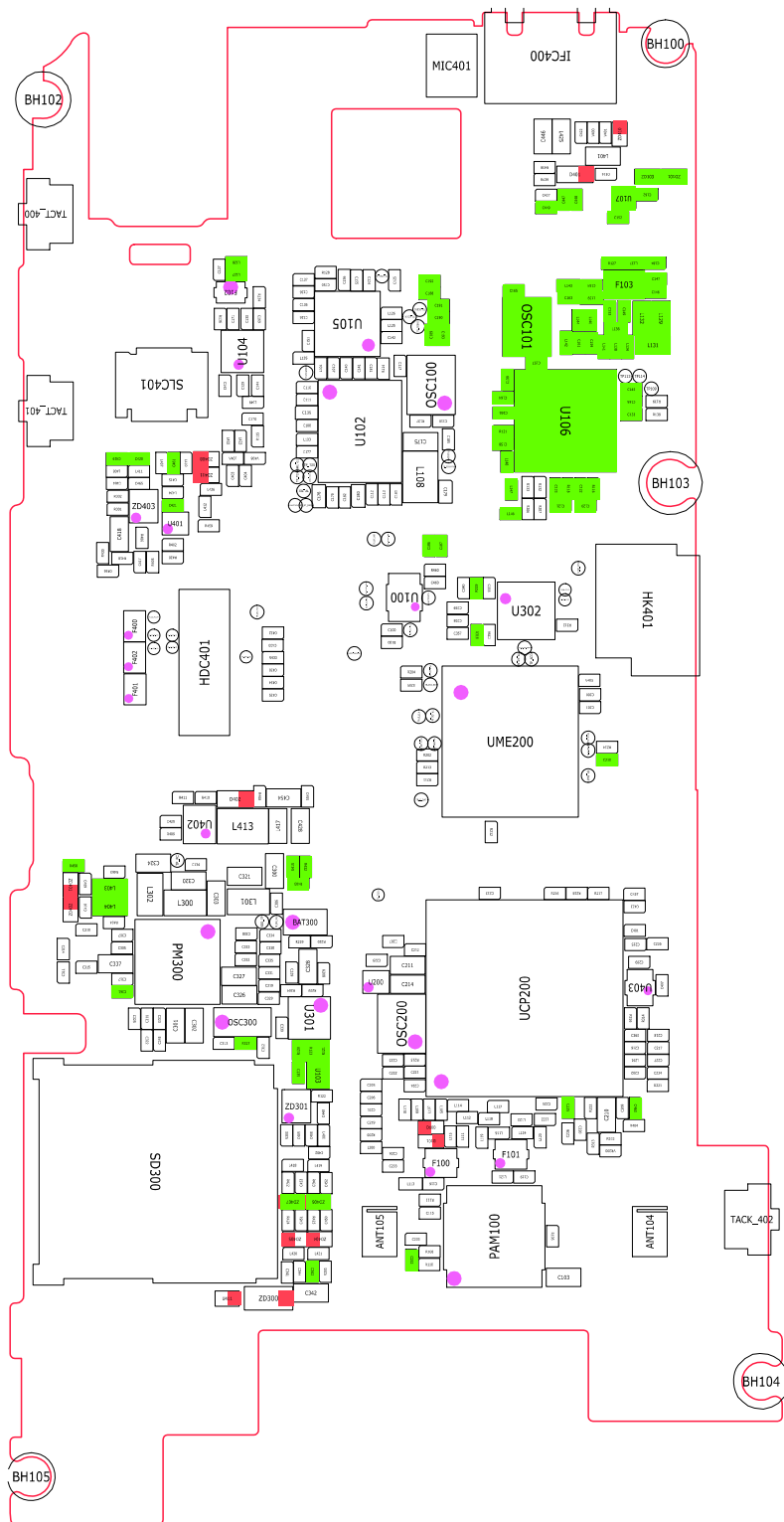
- BCM21334 (BASEBAND):** The central processor, connected to **SDDAT2-3, CAM, SD2CMD, CAM, SD2CLK, CAM**, **CAM_SCL, TV_RESET, CAM_SDA, CAM_VSYN, Y(2-7)**, **FM_ATV_EN, FM_ANT_EN**, **MAIN_MIC_SEL, MIC_P**, **AUX_MIC_P**, **SEL_REF_MIC**, **CAM_EN, CAM_RST, CAM_PCLK**, **SD2CMD, SD2CLK, KSD2DAT2-3**, **WIFI_SEL**, **BT_WAKE, HOST_WAKE, BT_PWR, BT_CLK, PCM, UART_RXD, TXD, RTS, CTS, 32K_CLK**, **SD2DAT2-3, WIFI, SD2CLK, WIFI, SD2CMD, WIFI**, **RST_WIFI, SD2DAT0, SD2DAT1**, **AMP_N, AMP_P**, **FM_SEL, EAR_L, EAR_R**, **VF**, **VDD_ANA_3.0V, VDD_CAM_A_2.8V, VDD_ANA_1.2V, VDD_VRF_1.3V, VDD_VRF_2.7V, VDD_USBOTG_2.5V, VDD_LCD_2.3V, VDD_USB_NVRAM_3.3V, VDD_CAM_IO_2.8V, VDD_CAM_CORE_1.2V, VDD_WOT_3.0V, VDD_TSP_3.0V, VDD_BT_3.0V, VDD_CORE_1.28V, VDD_MEM_1.8V, VDD_TV_1.2V, VDD_AMP1_3.0V, VDD_SD_3.2V, VDD_DIG_3.0V, VDD_VCO_1.3V**, **32K_CLK**, **RXD1**, **SCL, LDO_EN4, LDO_EN5, PC**, **SDA, USB_DM, USB_DP**.
- ANALOG SW:** Connected to **FM_ATV_EN, FM_ANT_EN**, **MAIN_MIC_SEL, MIC_P**, **AUX_MIC_P**, **SEL_REF_MIC**.
- FSA2466 (WIFI/CAM SW):** Connected to **SD2CMD, SD2CLK, KSD2DAT2-3**, **WIFI_SEL**.
- BCM20780 (BT/FM RADIO):** Connected to **BT_WAKE, HOST_WAKE, BT_PWR, BT_CLK, PCM, UART_RXD, TXD, RTS, CTS, 32K_CLK**, **SD2DAT2-3, WIFI, SD2CLK, WIFI, SD2CMD, WIFI**, **RST_WIFI, SD2DAT0, SD2DAT1**.
- BCM4319 (WIFI):** Connected to **SD2DAT2-3, WIFI, SD2CLK, WIFI, SD2CMD, WIFI**, **RST_WIFI, SD2DAT0, SD2DAT1**.
- AUDIO SW:** Connected to **FM_SEL, EAR_L, EAR_R**.
- MAX8959 (PMIC):** Connected to **VDD_ANA_3.0V, VDD_CAM_A_2.8V, VDD_ANA_1.2V, VDD_VRF_1.3V, VDD_VRF_2.7V, VDD_USBOTG_2.5V, VDD_LCD_2.3V, VDD_USB_NVRAM_3.3V, VDD_CAM_IO_2.8V, VDD_CAM_CORE_1.2V, VDD_WOT_3.0V, VDD_TSP_3.0V, VDD_BT_3.0V, VDD_CORE_1.28V, VDD_MEM_1.8V, VDD_TV_1.2V, VDD_AMP1_3.0V, VDD_SD_3.2V, VDD_DIG_3.0V, VDD_VCO_1.3V**, **32K_CLK**, **RXD1**, **SCL, LDO_EN4, LDO_EN5, PC**, **SDA, USB_DM, USB_DP**.

RF/IO (Right Section, Light Orange):

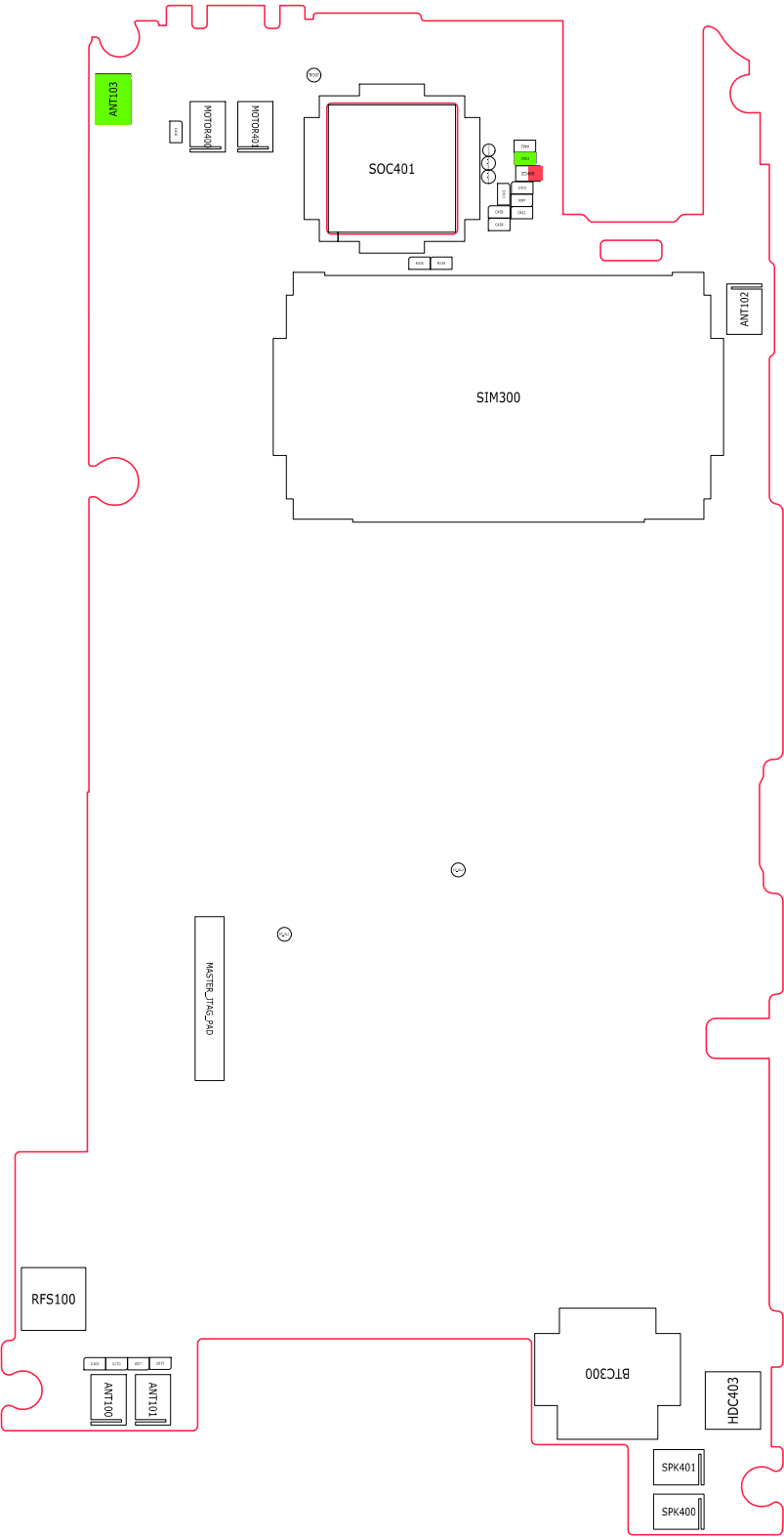
- TLG1121 (A-TV):** Connected to **VDD_TV_1.2V, VDD_CAM_IO_2.8V**, **ATV_ANT**, **ATV_ANT_FM**, **TV_AUDIO_R, TV_AUDIO_L**, **TV_AUDIO_L**, **MAIN_MIC_P**, **SUB_MIC**, **3M CAM**, **VDD_DIG_3.0V**, **VDD_BT_3.0V, VREG_1.2V, VDD_DIG_3.0V**, **FM_ANT_SW**, **BT_REQ_CLK**, **FM_LOUT, ROUT**, **BT_RF**, **COEX(0-3)**, **VBAT, VDD_DIG_3.0V**, **VREG_PLL_1.2V**, **WIFI_RX**, **EB_BT_TX, EB_WIFI_TX, EN_WIFI_RX**.
- S/W (ATV/FM):** Connected to **ATV_ANT**, **ATV_ANT_FM**, **FM_ANT**.
- 3M CAM:** Connected to **VDD_DIG_3.0V**.
- FET:** Connected to **VDD_DIG_3.0V**, **3.5PI CON DET**.
- SW (WIFI/BT):** Connected to **WIFI_RX**, **EB_BT_TX, EB_WIFI_TX, EN_WIFI_RX**.
- 3.5PI Ear Jack:** Connected to **FM_ANT**.
- WIFI_TX:** Connected to **WIFI_TX**.
- MAX8959 (PMIC):** Connected to **VDD_ANA_3.0V, VDD_CAM_A_2.8V, VDD_ANA_1.2V, VDD_VRF_1.3V, VDD_VRF_2.7V, VDD_USBOTG_2.5V, VDD_LCD_2.3V, VDD_USB_NVRAM_3.3V, VDD_CAM_IO_2.8V, VDD_CAM_CORE_1.2V, VDD_WOT_3.0V, VDD_TSP_3.0V, VDD_BT_3.0V, VDD_CORE_1.28V, VDD_MEM_1.8V, VDD_TV_1.2V, VDD_AMP1_3.0V, VDD_SD_3.2V, VDD_DIG_3.0V, VDD_VCO_1.3V**, **32K_CLK**, **RXD1**, **SCL, LDO_EN4, LDO_EN5, PC**, **SDA, USB_DM, USB_DP**.
- IFCON MicroUSB:** Connected to **VBAT, VBUS**, **USB_D+, USB_D-**.

8-2.主板图

8-2-1.顶部



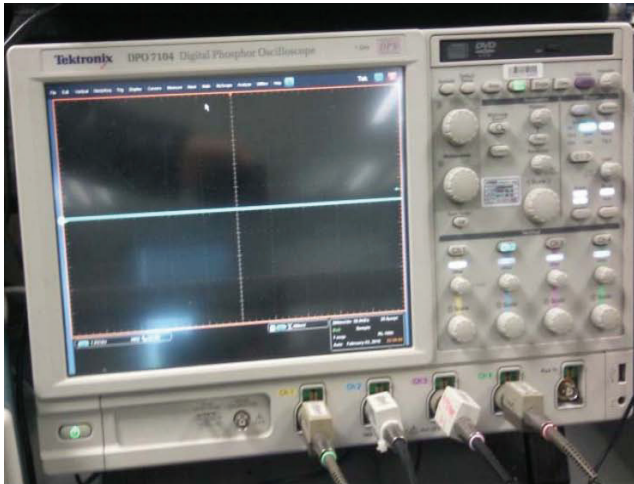
8-2-2.底部



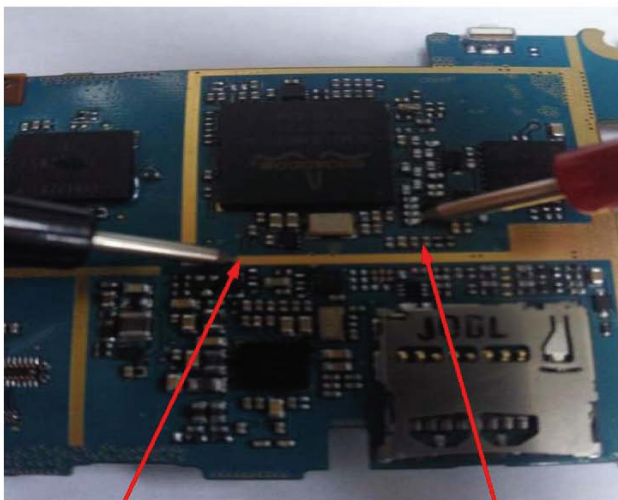
8-3.故障排除流程图

※测试点检查的预调整方法

- GND 和 TP（例如：VBAT=C308）使用示波器
- 检查输出信号



←示波器

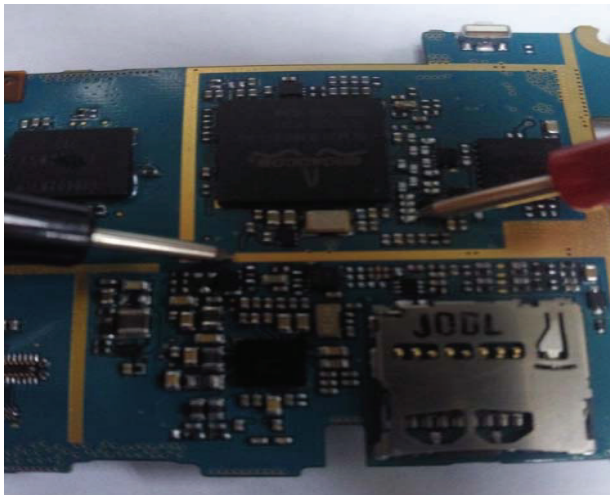


GND

TP (测试点)



←多功能仪表

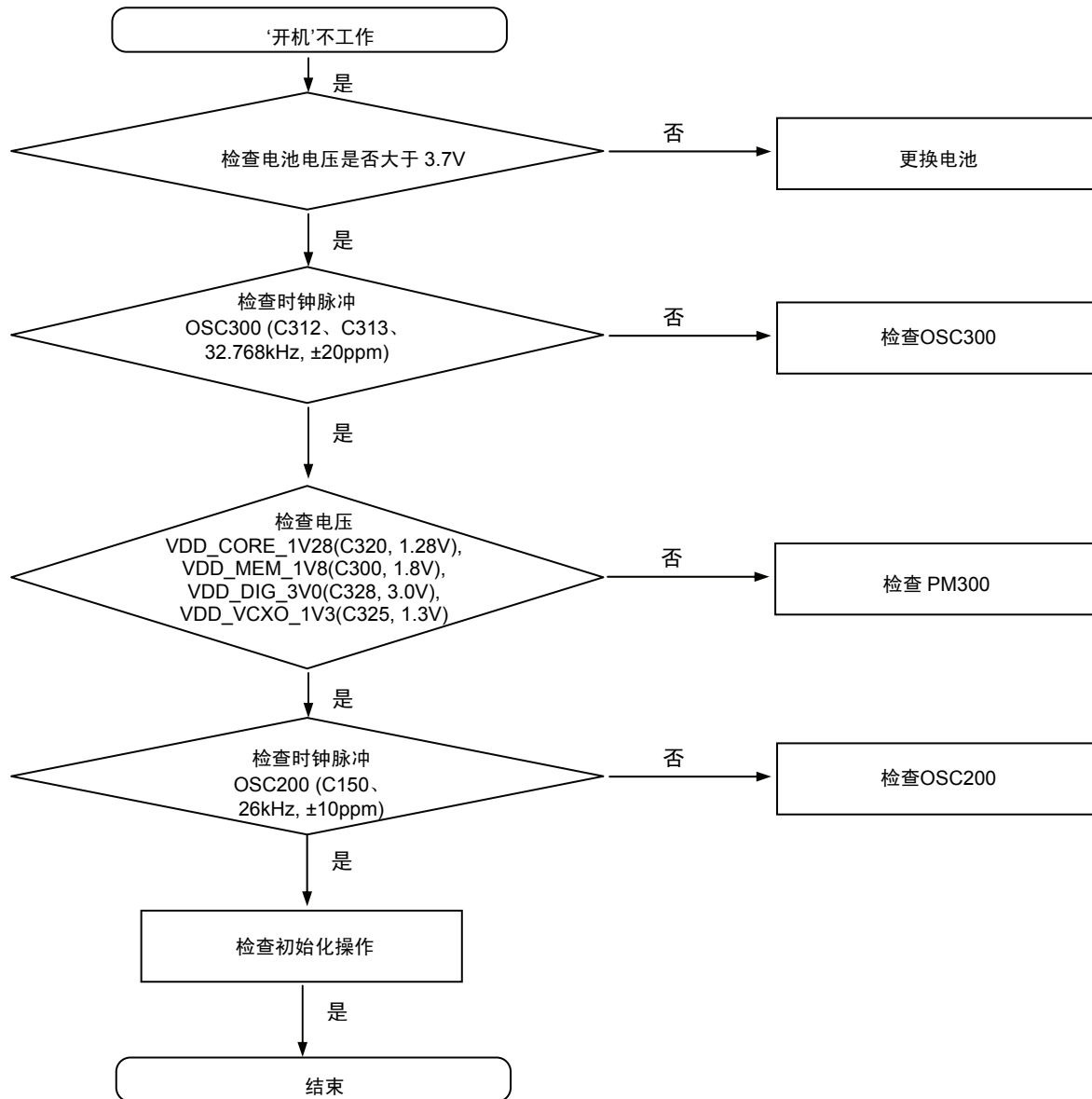


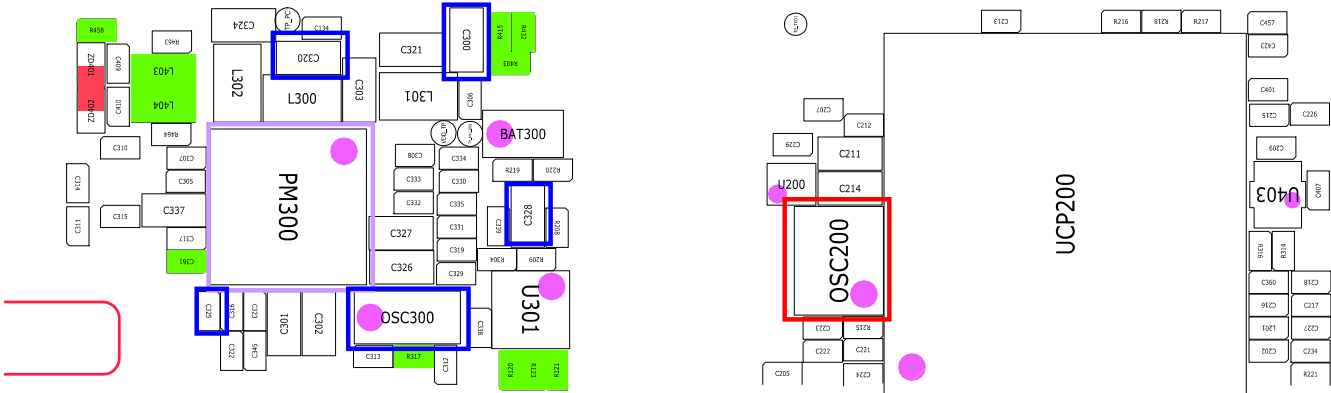
←使用多功能仪表检查 TP（测试点）

- 例如）检查测试点，分路电容。
- 如果检查 GND，你可以听到“哔哔声”。
- 如果检查信号，你将不会听到有声音发出。

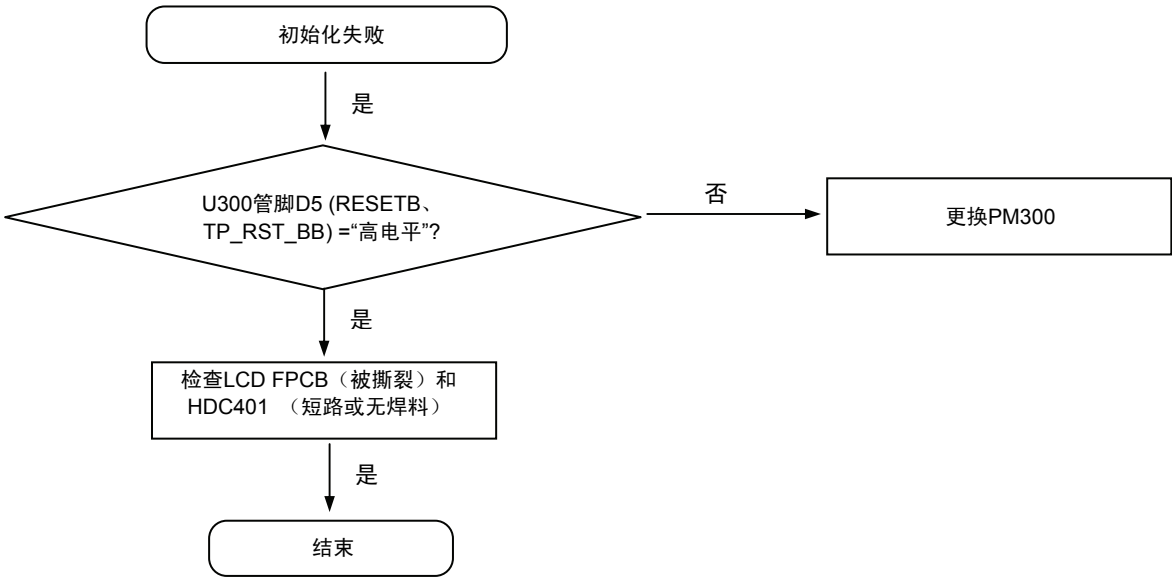
8-3. 逻辑

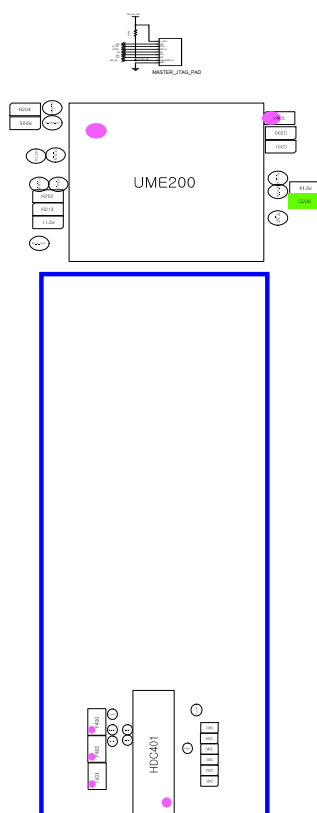
8-1. 开机



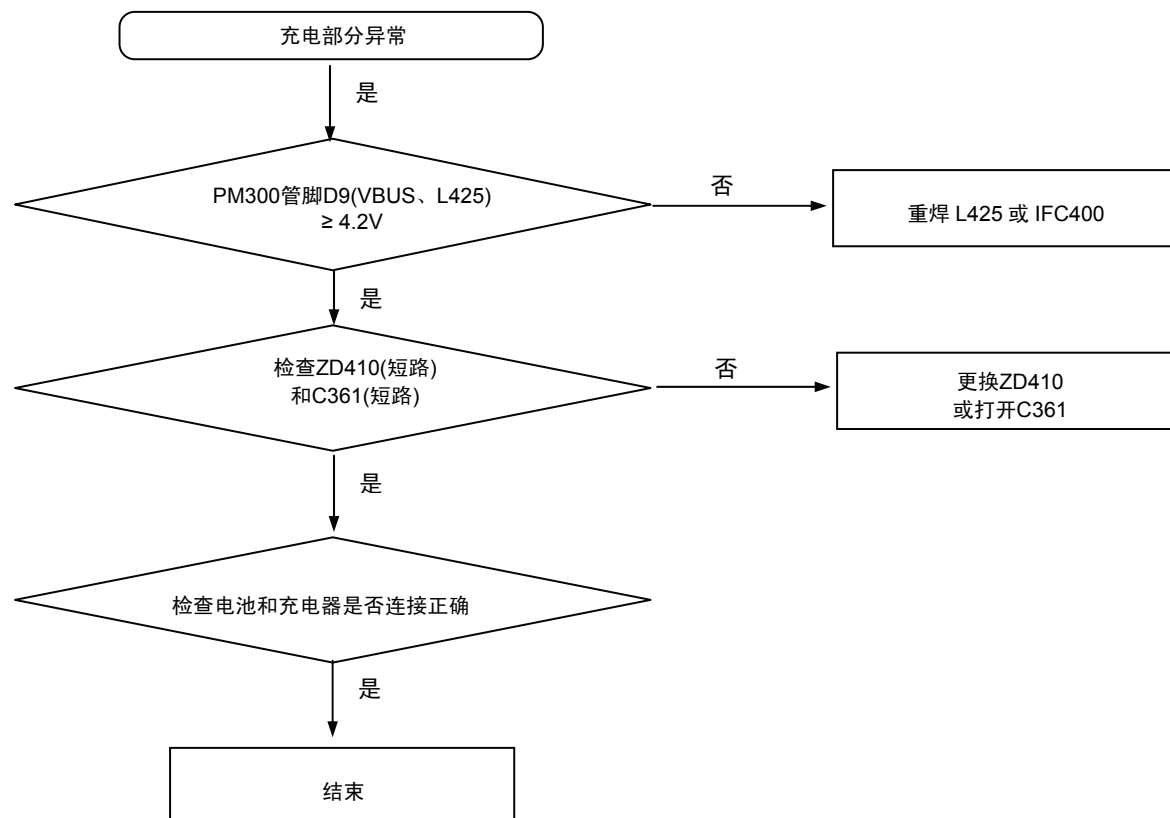


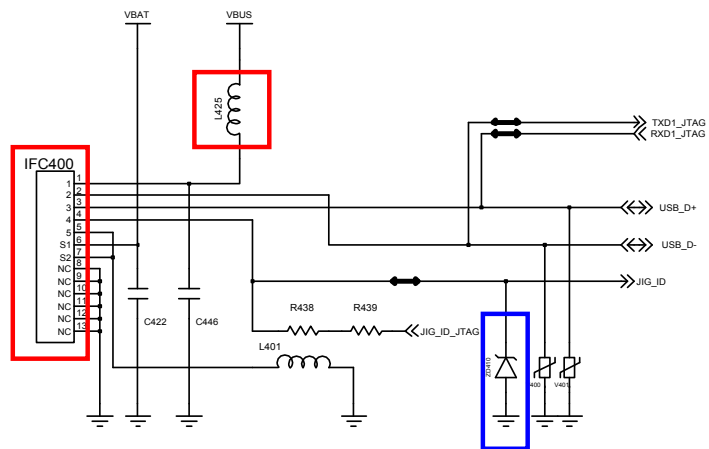
8-2. 初始化



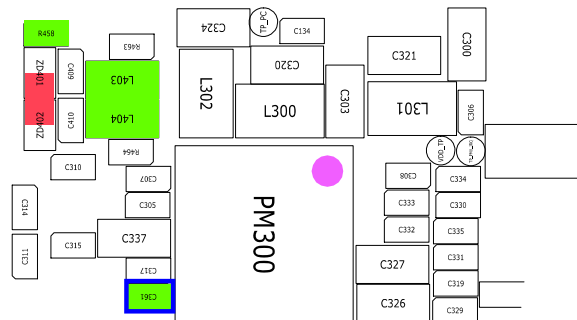
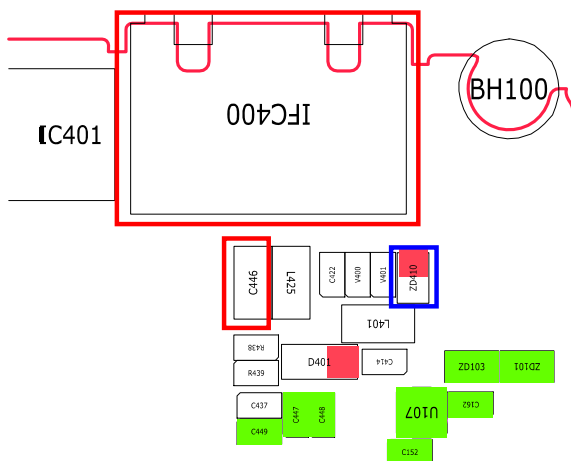
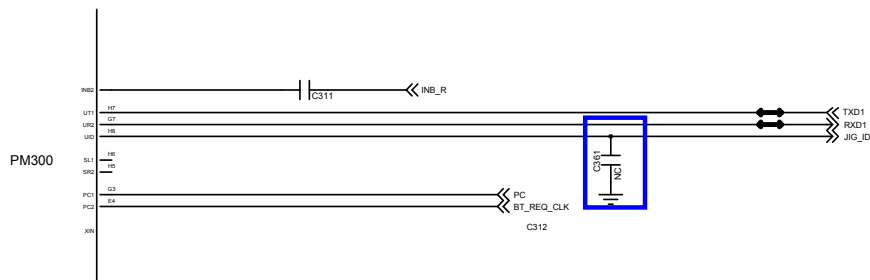


8-3. 充电部分

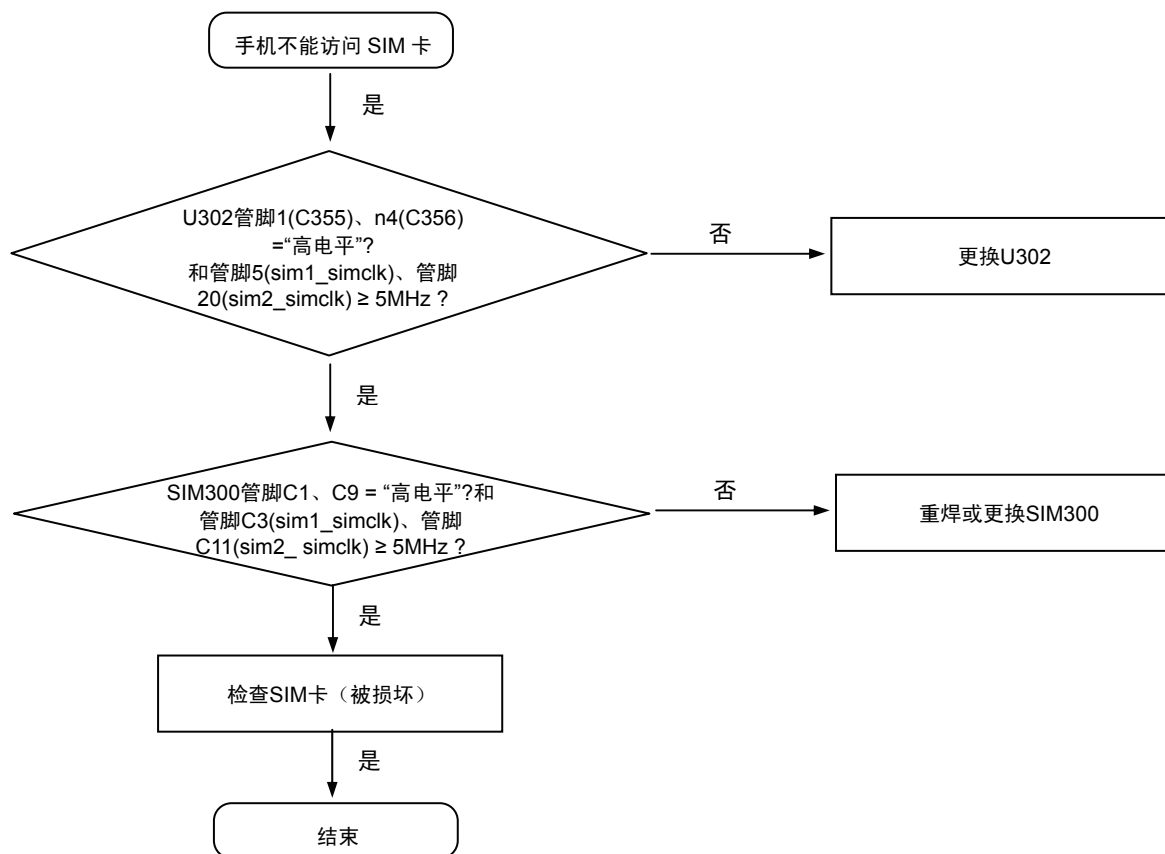


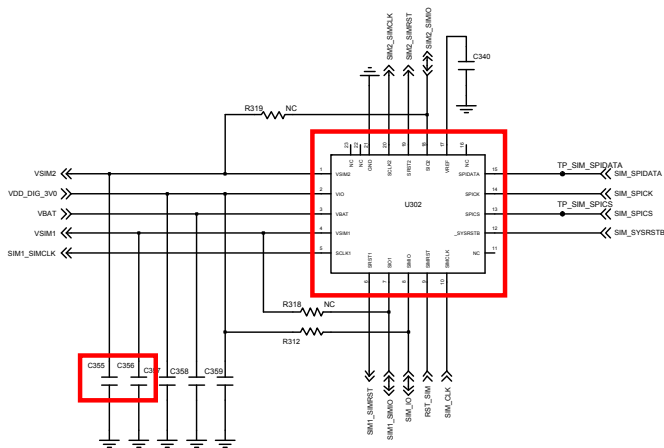


IF CON

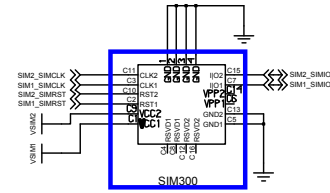


8-4. SIM 部分

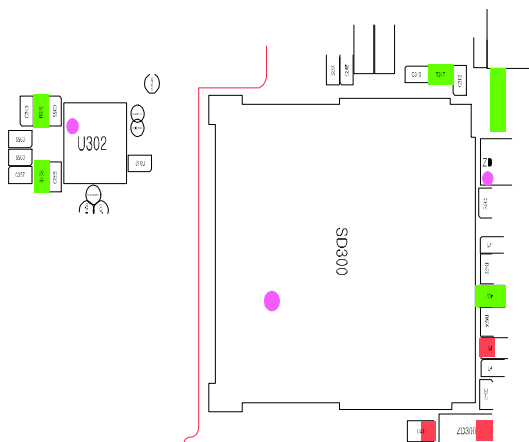




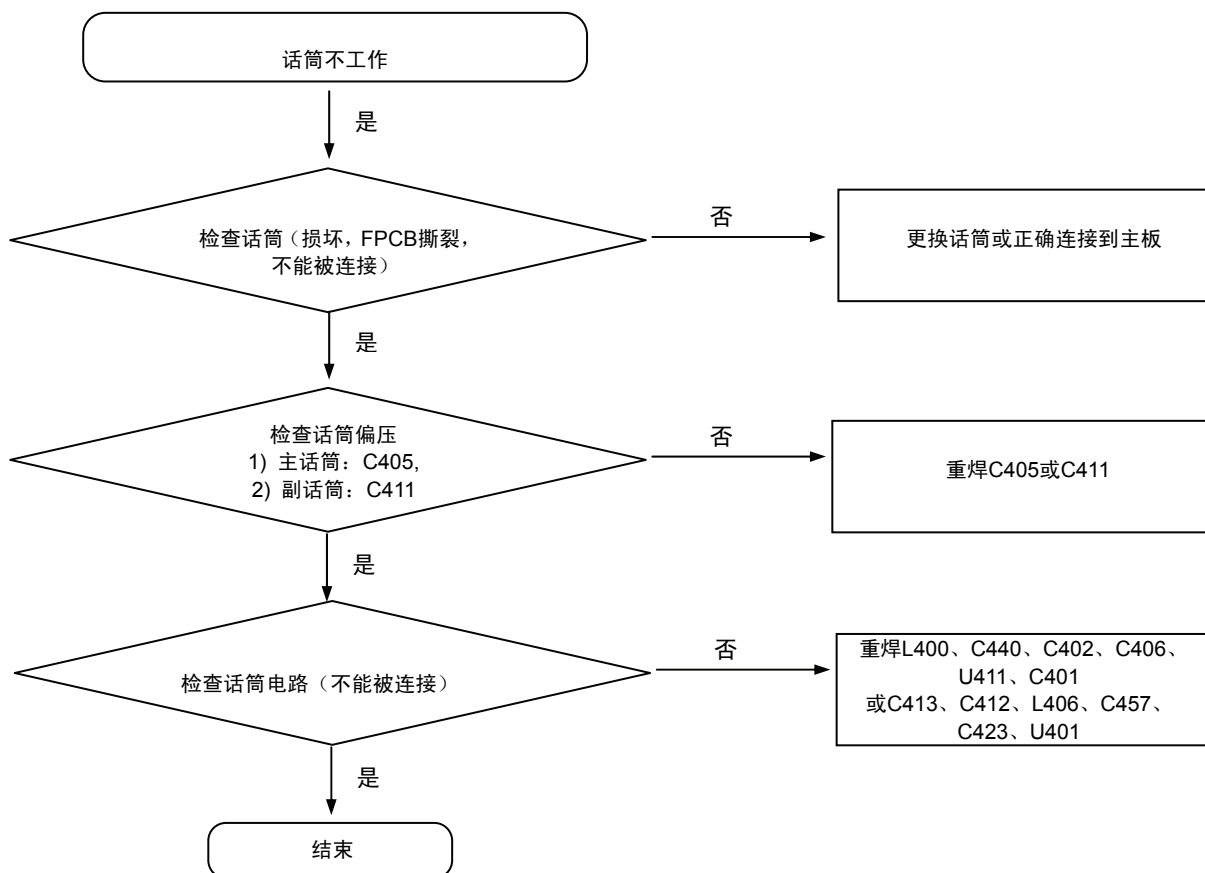
DUAL SIM CONTROLLER

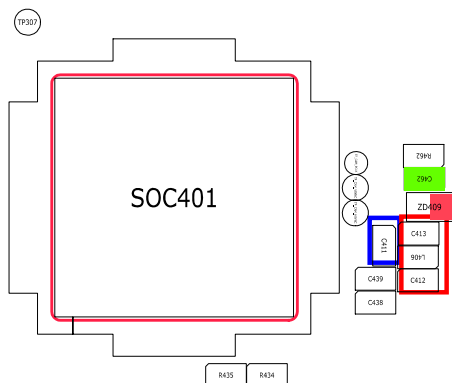
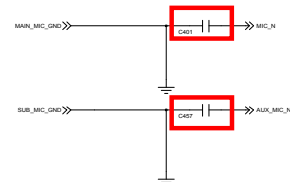
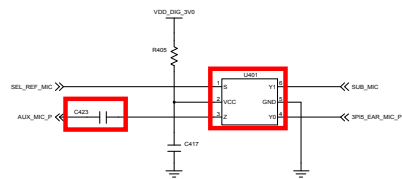
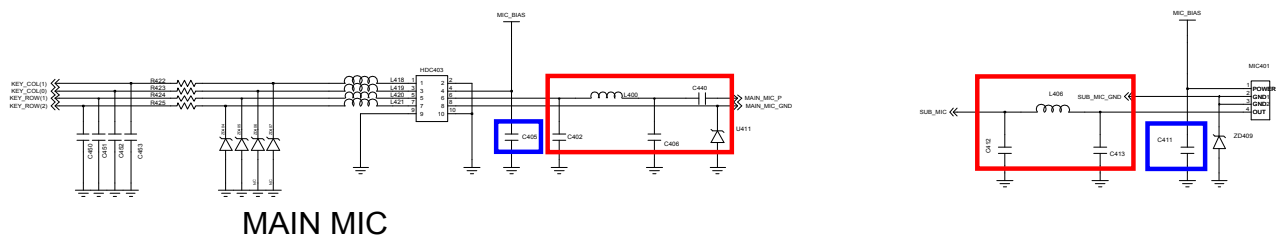


DUAL SIM CONNECTOR

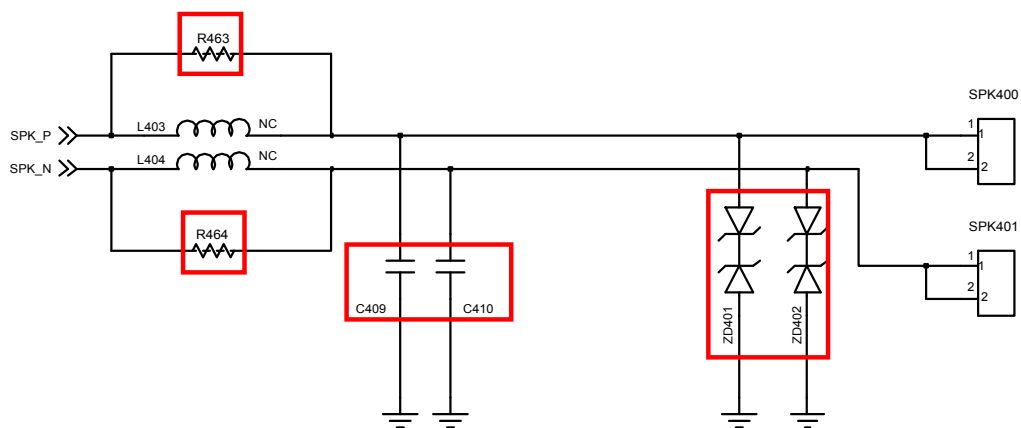
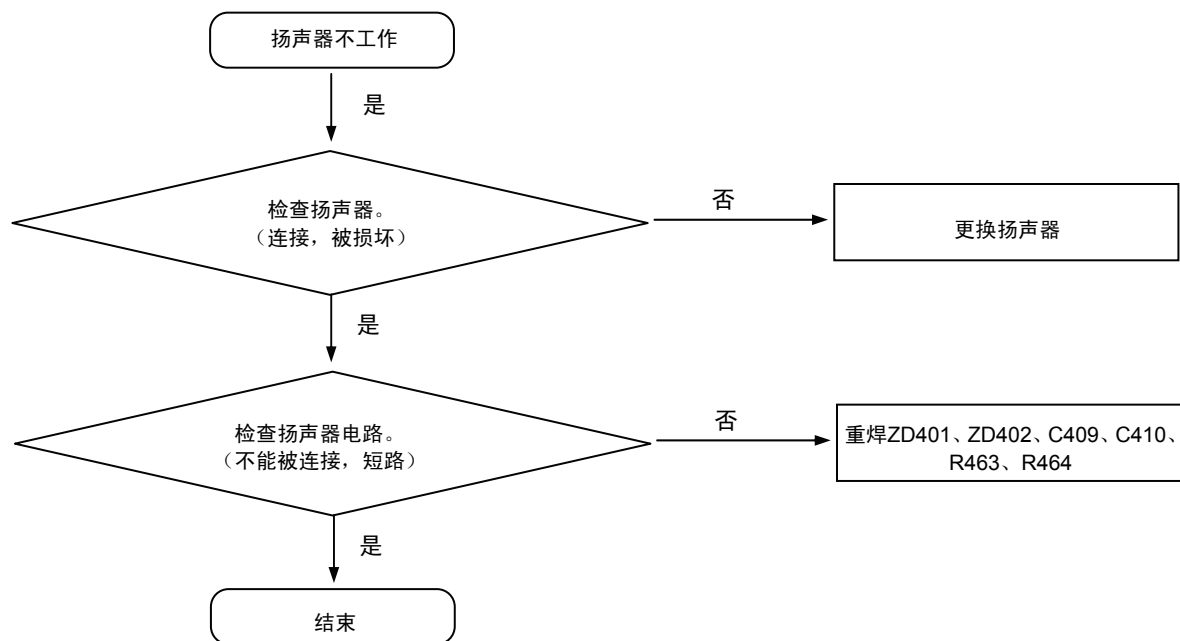


8-5. 话筒部分

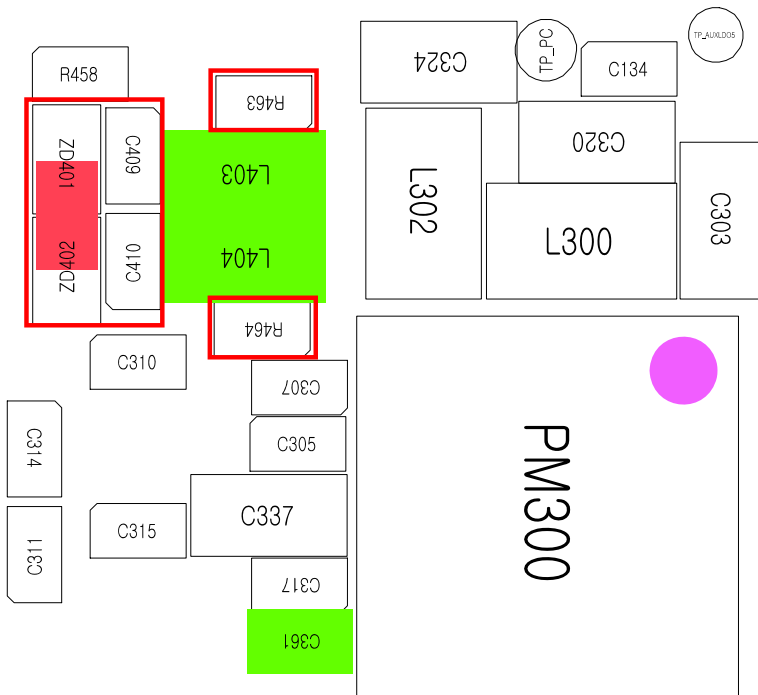




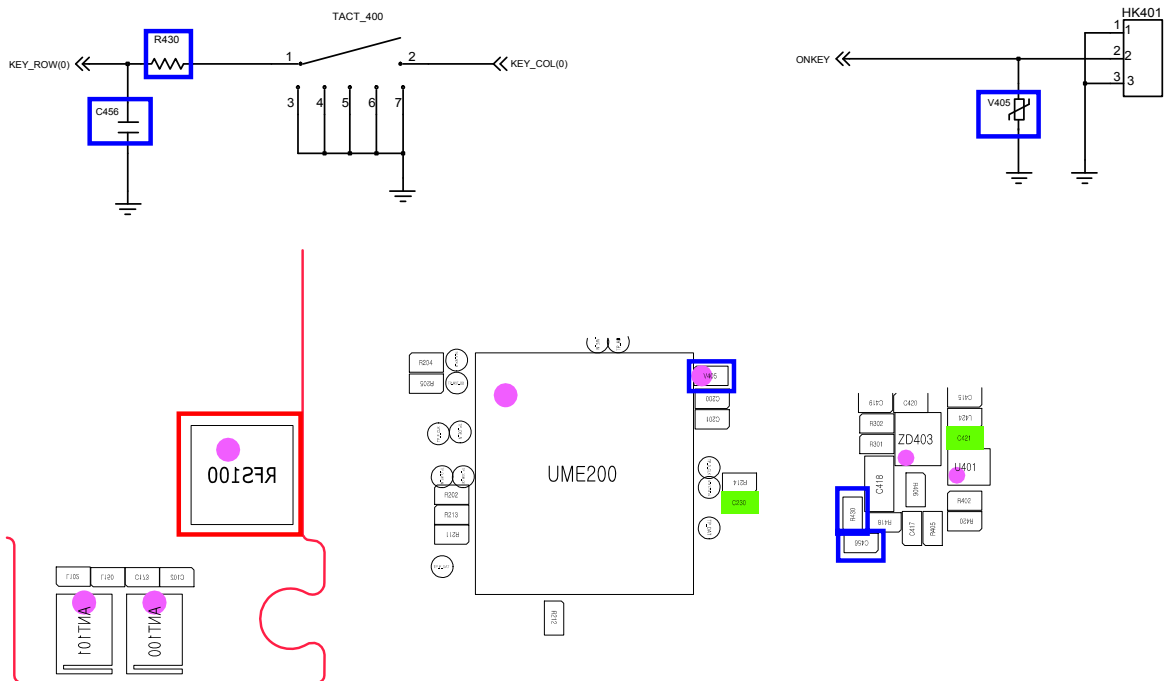
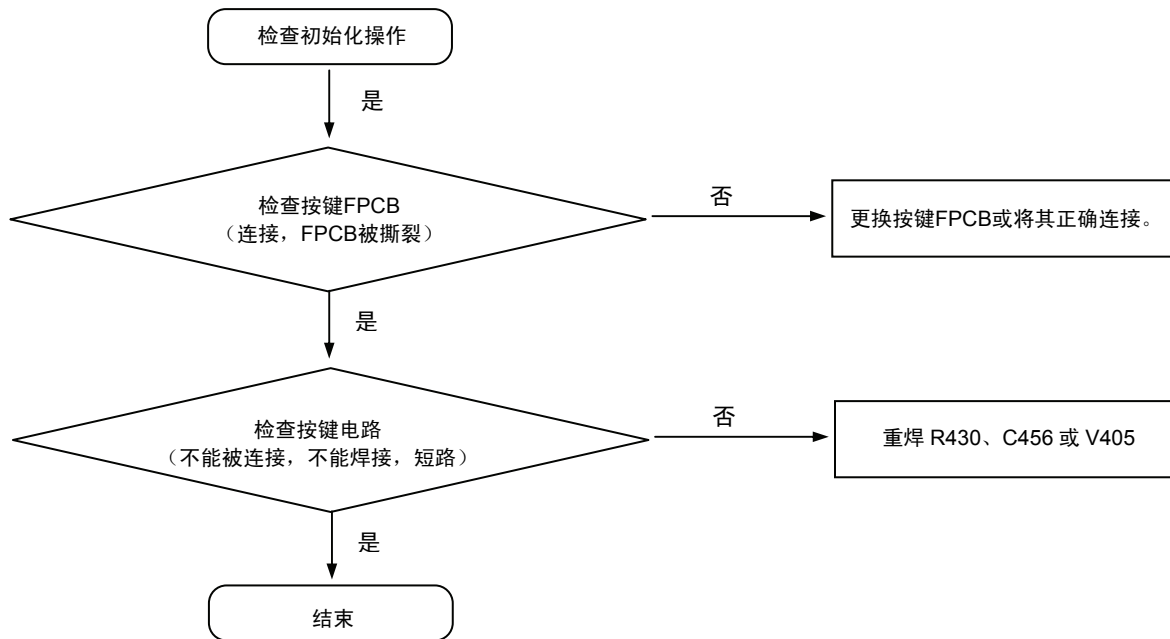
8-6. 扬声器部分



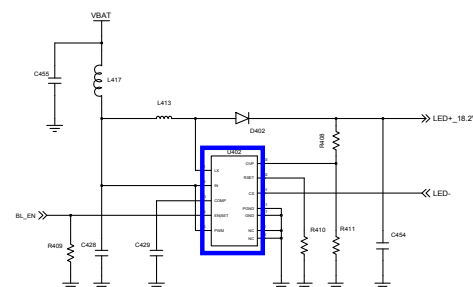
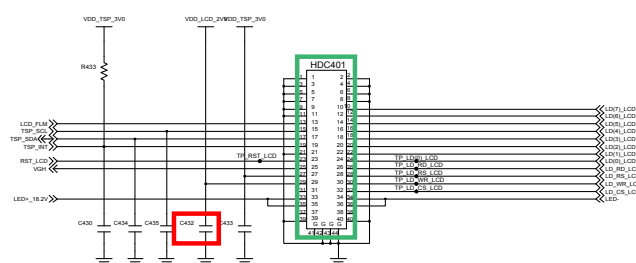
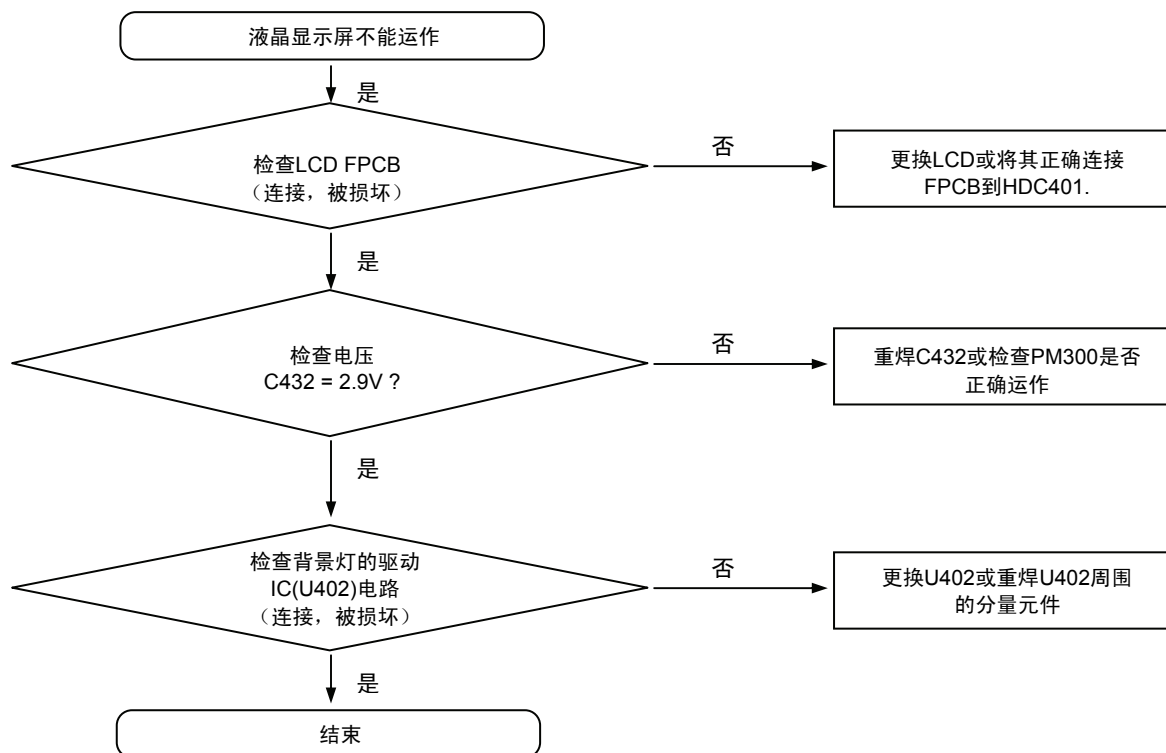
SPK PAD

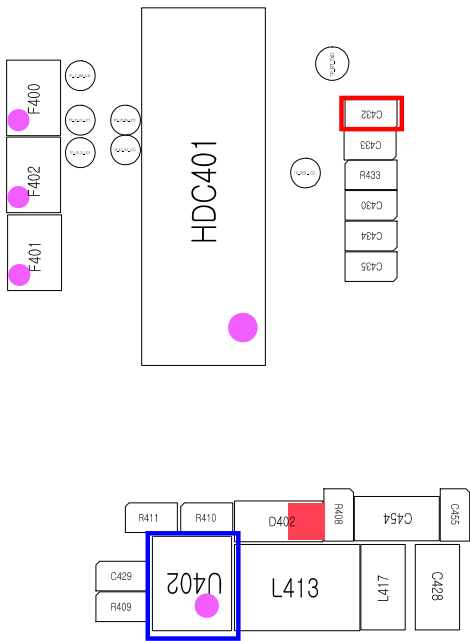


8-7.按键数据输入

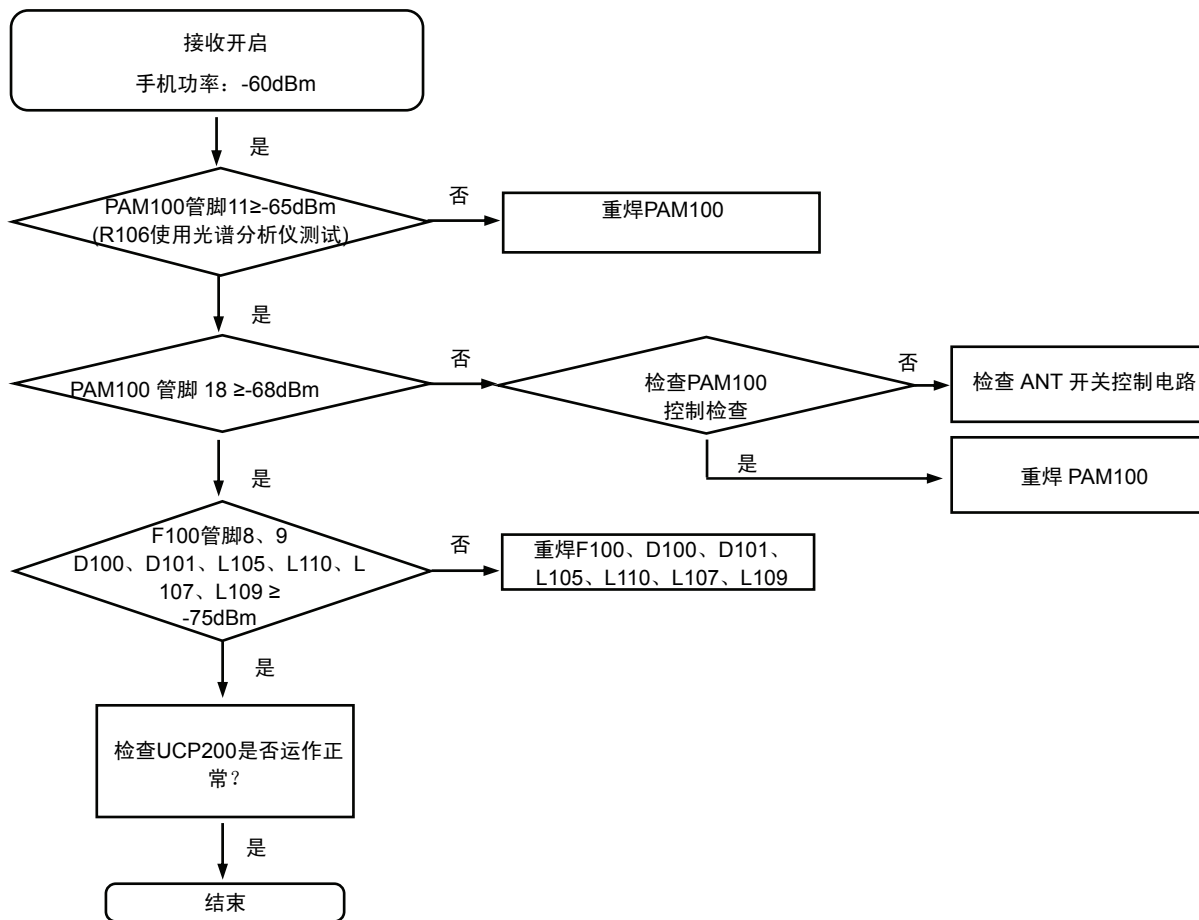


8-8.背景灯（液晶显示屏）

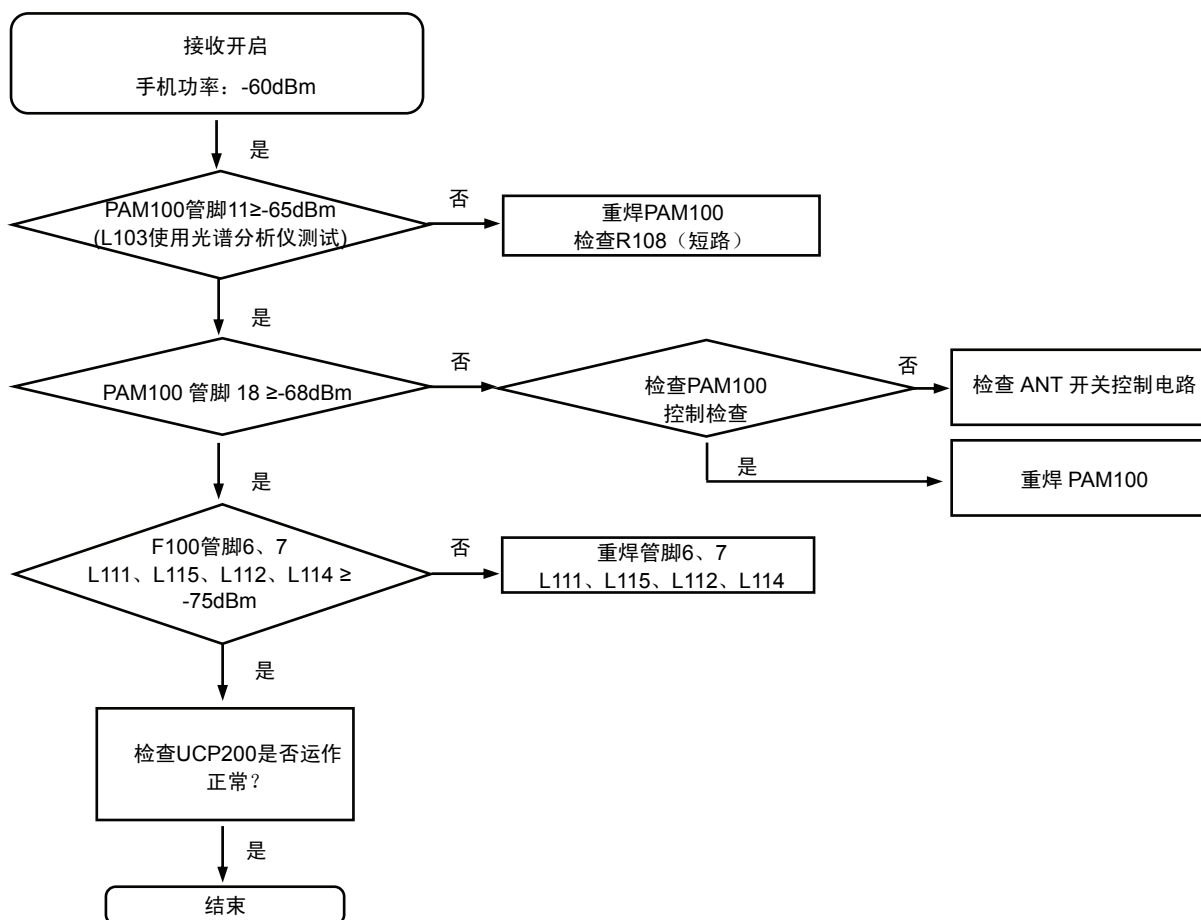




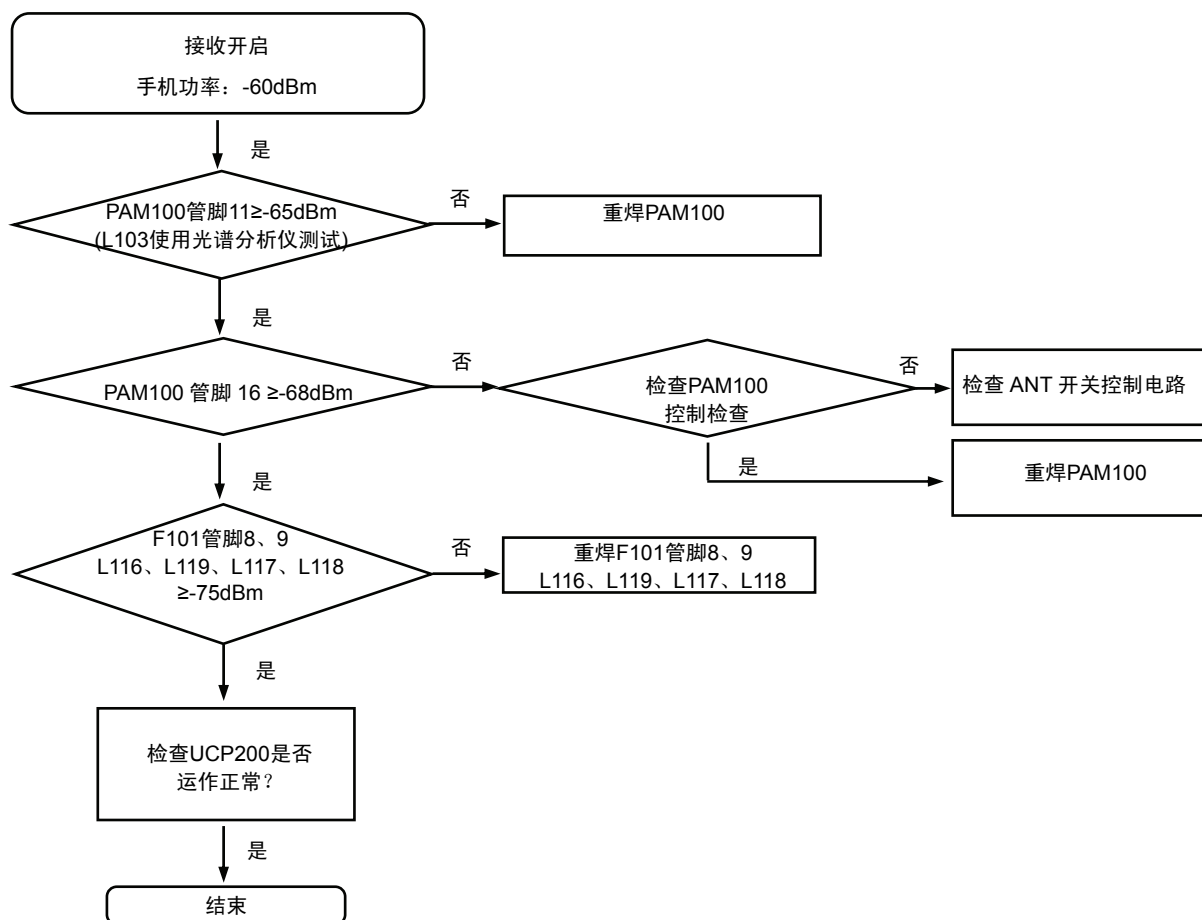
8-9.GSM850 接收器



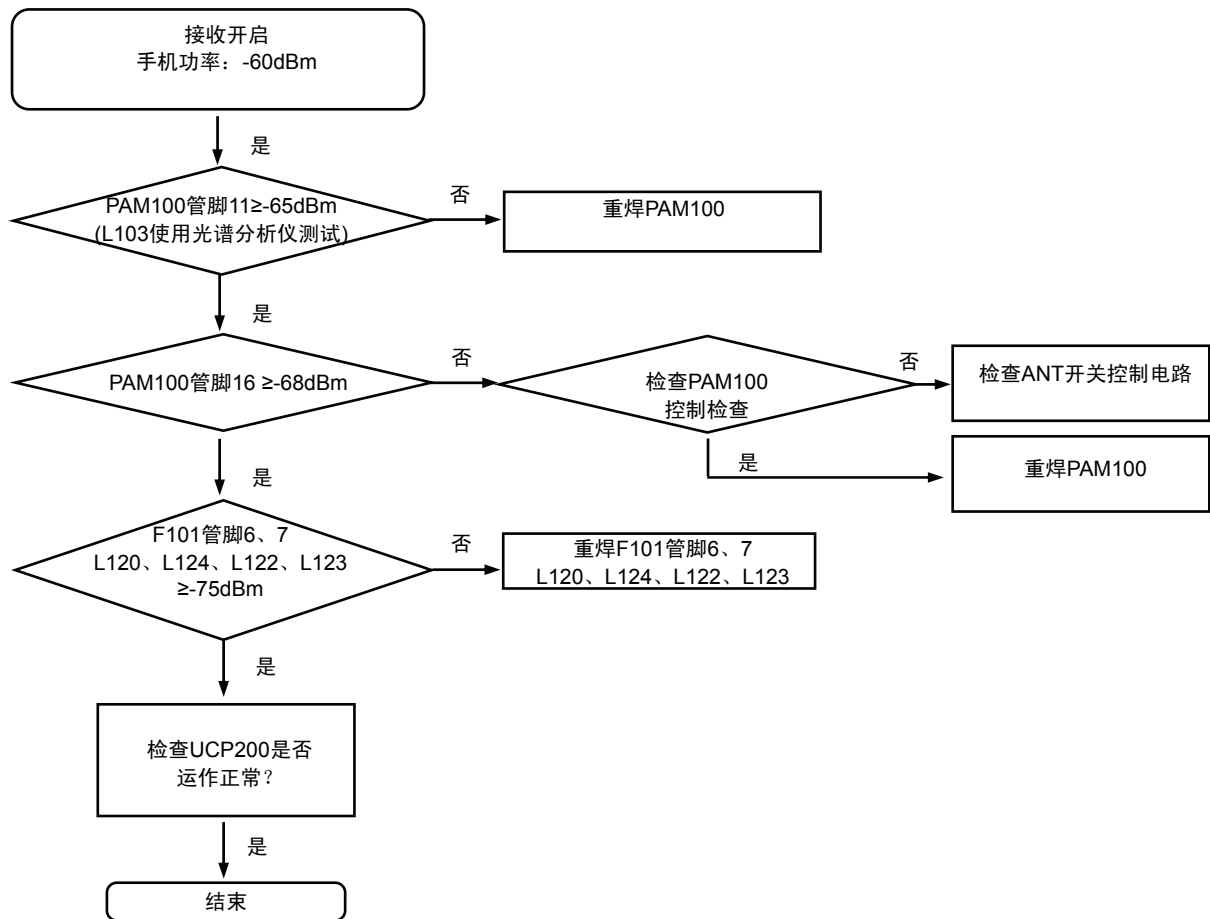
8-10.GSM900 接收器

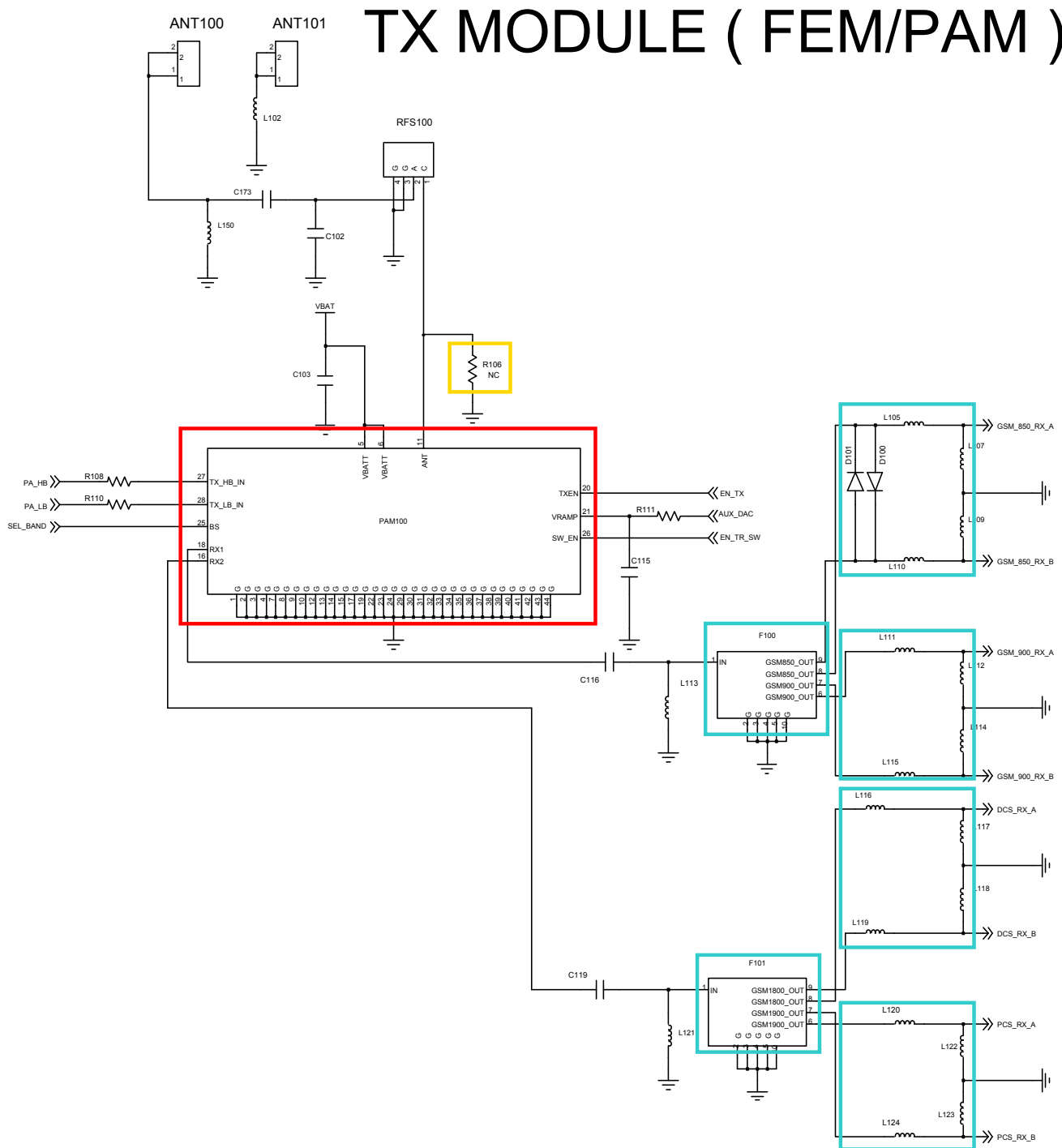


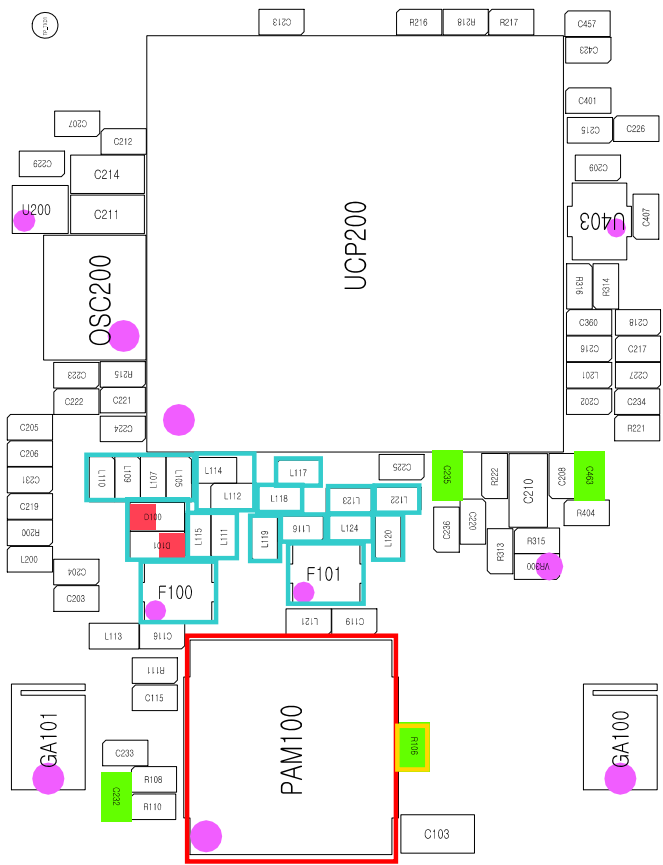
8-11.DCS 接收器



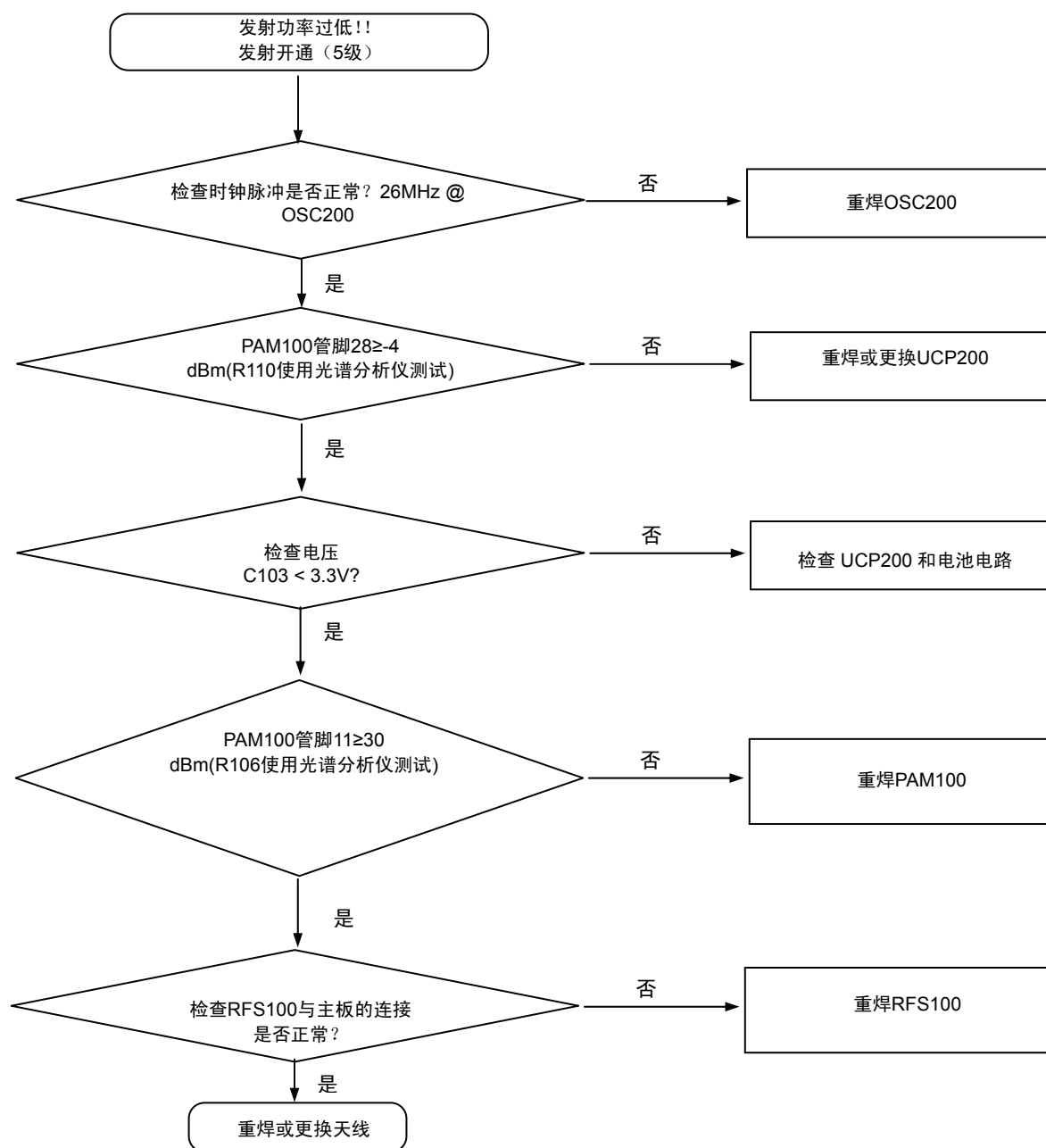
8-12.PCS 接收器



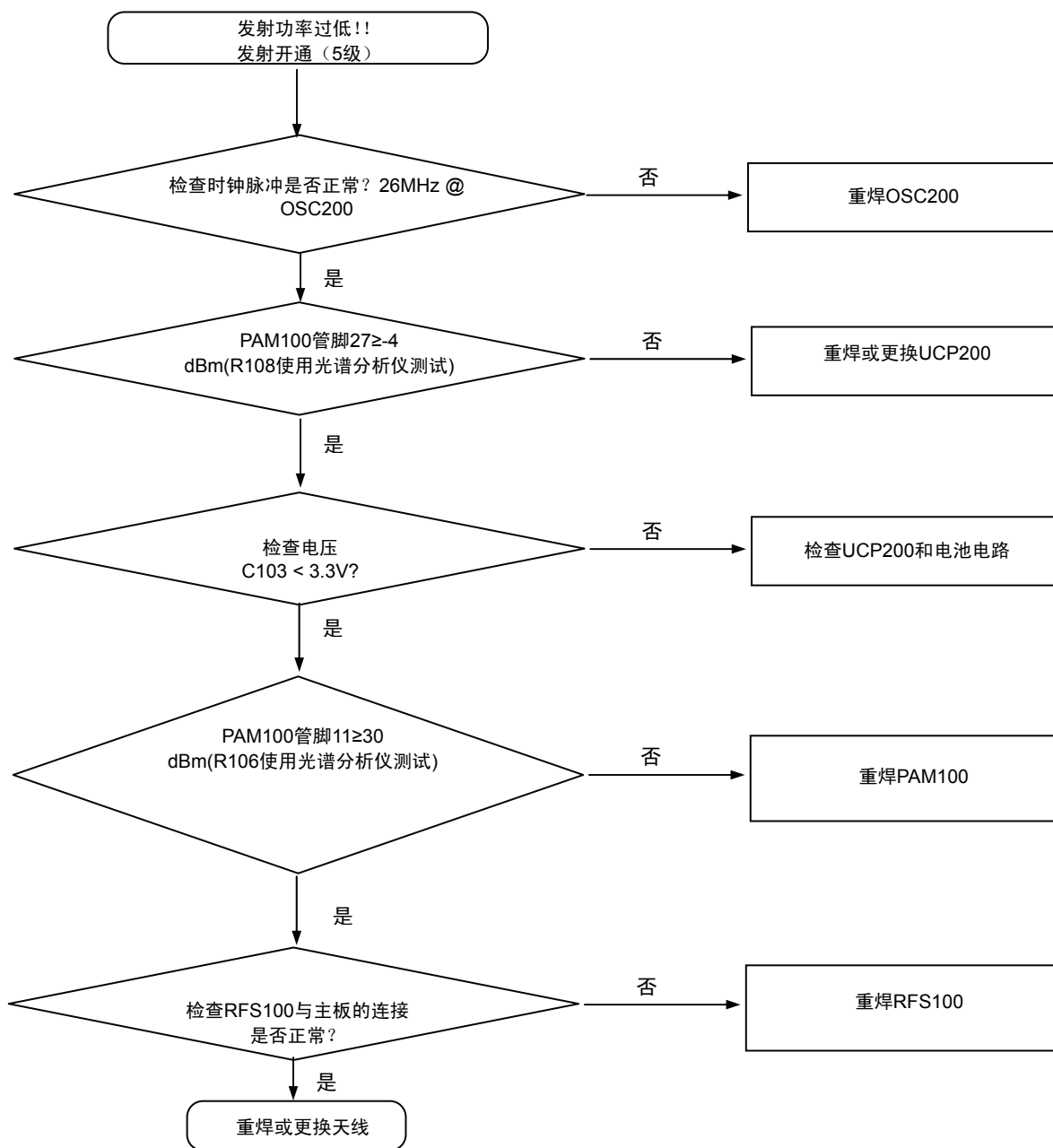


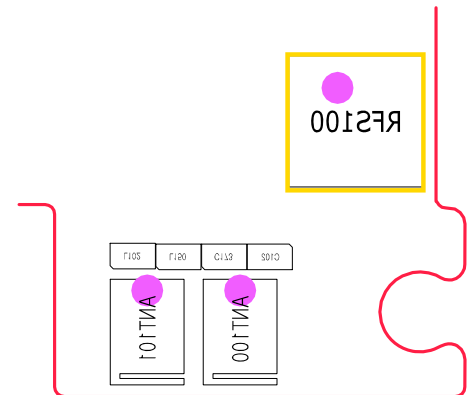
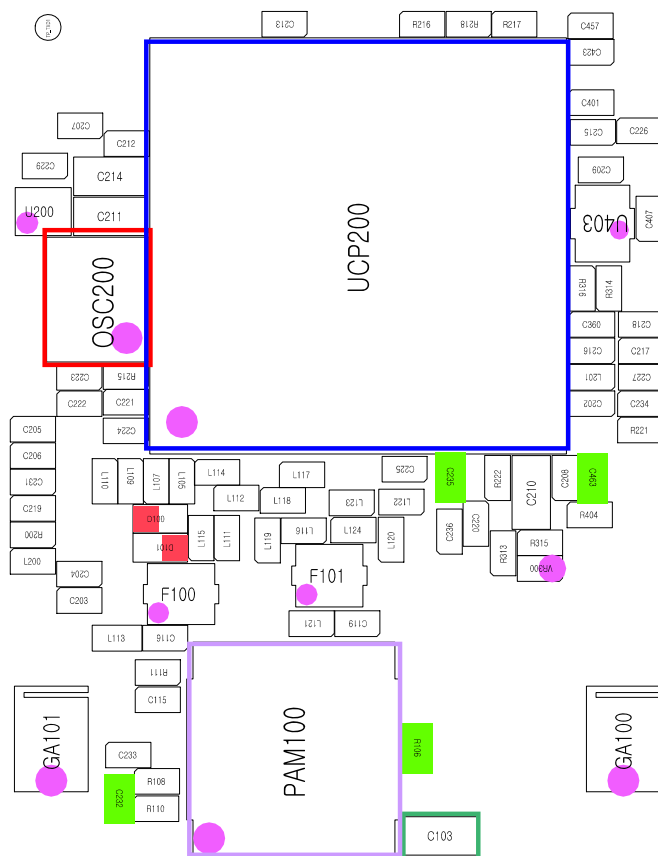
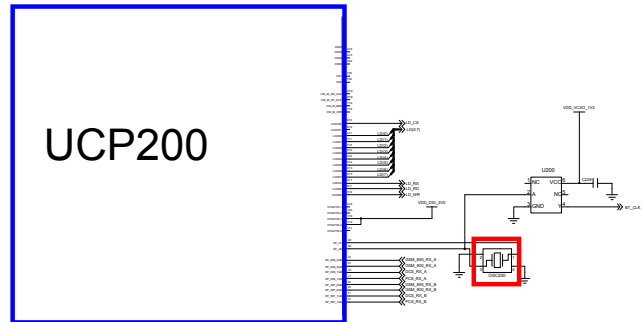
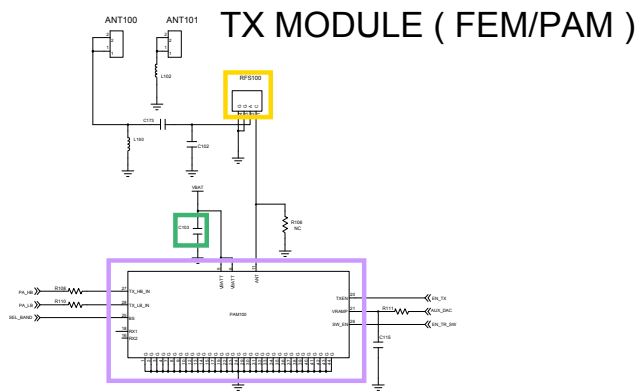


8-13.GSM850/900 发射器



8-13.DCS/PCS 发射器





-故障处（顶部视图）

● : NC

U300

	1	2	3	4	5	6	7	8	9
A	○	○	○	○	○	○	○	○	○
B	○	○	○	○	○	○	○	○	○
C	○	○	○	○	○	○	○	○	○
D	○	○	○	○	○	○	○	○	○
E	○	○	○	○	○	○	○	○	○
F	○	○	○	○	○	○	○	○	○
G	○	○	○	○	○	○	○	○	○
H	○	○	○	○	○	○	○	○	○
J	○	○	○	○	○	○	○	○	○

UME200

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	DNU	DNU	NC	VSSo	VCCo	VSSQd	VDDQd	VDDQd	VSSQd	VSSd	VDDd	VSSQd	DNU	DNU
B	DNU	VSSo	/OEo	NC	/RPe	/WEo	NC	NC	NC	DQ12d	NC	NC	VDDQd	DNU
C	VSSd	NC	/WEd	AVDo	/CEo	RDYo	DQ14d	DQ8d	DQ13d	NC	NC	DQ9d	UDMd	VDDQd
D	VDDd	/CSd	BA0d	Index								NC	NC	VSSQd
E	NC	/RASd	A2d		VCCo	NC	NC	INTo	NC	NC		NC	DQ15d	UDQ3d
F	/CASd	A12d	A0d		CLKo					NC		DQ11d	DQ10d	VSSQd
G	CKEd	A9d	BA1d		VSSo					NC		VDDd	VDDQd	CKd
H	VDDd	A11d	A7d		ADQ8o					ADQ15o		VSSd	VDDQd	/CKd
J	A4d	VSSd	A5d		ADQ9o					ADQ14o		LDQ3d	NC	VSSQd
K	A6d	A10d	A3d		ADQ10o	ADQ11o	VCCo	VSSo	ADQ12o	ADQ13o		DQ2d	LDMd	DQ4d
L	A13d	A8d	A1d									DQ5d	DQ7d	VSSQd
M	VSSd	VDDd	NC	ADQ5o	ADQ2o	ADQ0o	DQ6d	DQ3d	NC	NC	NC	NC	DQ0d	VDDQd
N	DNU	VCCo	NC	ADQ6o	ADQ3o	VSSQd	NC	DQ1d	NC	NC	NC	NC	VDDQd	DNU
P	DNU	DNU	VSSo	ADQ7o	ADQ4o	ADQ1o	VDDQd	VDDQd	VSSQd	VSSd	VDDd	VSSQd	DNU	DNU

153 FBGA: Top View (Ball Down)

	OneNAND Flash
	Mobile DDR SDRAM
	Power
	Ground
	NC/DNU

UCP200

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
A	○	○	●	●	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
B	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
C	○	○																				○	○
D	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
E	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
F	○	○		○	○	○	○	●	○	○	○	○	○	○	○	○	○	○	○	○		○	○
G	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
H	○	○		○	○	○	○	○	○	○	○	○	○	○			○	○	○	○		○	○
J	○	○		○	○	○	○										○	○	○	○		○	○
K	○	○		○	○	○	○										○	○	○	○		○	○
L	○	○		○	○	○	○			○	○	○	○	○			○	○	○	○		○	○
M	○	○		○	○	○				○	○	○	○	○			○	○	○	○		○	○
N	○	○		○	○	○	○			○	○	○	○	○			○	○	○	○		○	○
P	○	○		○	○	○	○			○	○	○	○	○			○	○	○	○		○	○
R	○	○		○	○	○	○										○	○	○	○		○	○
T	○	○	●	○	○	○	○										○	○	○	○		○	○
U	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
V	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
W	○	●		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
Y	○	●		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○
AA	○	○	○																	○	○	○	○
AB	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	●	○	○	○	○	○	○
AC	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	●	○	○	○	○	○	○